## Affiliation
Instituto de Fisica Teorica (IFT), CSIC

## Collaborations
COMPACT
Euclid
LIGO
VIRGO
LISA

## Website
**georgealestas.github.io**

## Tel & Skype
(+30) 6989964269
alestasg

## E-mail
**alestasg@gmail.com**

## Languages
**Greek** → Native
**English** → Fluent


# Dr. GeorgeAlestas


Physicist - Researcher


## Professional Summary

A highly motivated researcher specializing in Cosmology and Astrophysics. I am proficient in the theoretical and numerical exploration of cosmological models, including cosmological parameter inference using Bayesian analysis and machine learning techniques. I have also worked on aspects of the Hubble and $S_8$ tensions. Part of the Euclid, COMPACT, LIGO, Virgo and LISA collaborations. I have co-supervised three MSc. and two BSc. students. I would describe myself as a fast learner and an excellent collaborator.

## Positions & Academic Visits

2023 **Visiting Researcher** — MIT & Harvard
Institute for Artificial Intelligence and Fundamental Interactions (IAIFI), Boston

2023 **Visiting Researcher** — Nagoya University
Kobayashi-Maskawa Institute (KMI), Japan

2022 - Now **Postdoctoral Studies** — Universidad Autónoma de Madrid
Instituto de Fisica Teorica (IFT), Madrid

2018 - 2022 **PhD in Cosmology** — University of Ioannina
*PhD Thesis:* "ΛCDM and the Implications of the Hubble Tension"
*Advisor:* Prof. Leandros Perivolaropoulos
Grade: Summa Cum Laude

## Scholarships & Grants

2022-23 **Grant for IFT - Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) exchange program on academic visits.**

2022-23 **Application of machine learning to new cosmological observations, Funding institution: Ministerio de Ciencia e Innovación, i-LINK 2021**

2021-22 **Fellow of the Greek State and the European Union (European Social Fund – ESF) through the Operational Programme "Competitiveness, Entrepreneurship and Innovation 2014-2020 (EPAnEK)"**

2020-21 **Fellow of the Greek State and the European Union (European Social Fund – ESF) through the Operational Programme "Human Resources Development, Education and Lifelong Learning 2014-2020"**

2020 **European Cooperation in Science & Technology (COST) grant for a short term scientific mission (STSM) in the context of the action "CA15117 - Cosmology and Astrophysics Network for Theoretical Advances and Training Actions (CANTATA)" and the project "Search for Hints of Modified Gravity in Cosmological Data"**

## Conferences Attended & Talks Given

- Invisibles25, CERN, September 2025, **(Invited Talk)**
- Seminar at the Smithsonian center for astrophysics, November 2023, Harvard, **(Talk)**
- Cosmo23 - 2023 Conference in Cosmology, September 2023, Madrid, Spain, **(Co-Organizer)**
- Kickoff workshop, June 2023, Nagoya University, Japan, **(Invited Talk)**
- i-Link workshop, June 2023, Nagoya University, Japan, **(Invited Talk)**
- Seminar at ICF, February 2023, UNAM, Mexico, **(Invited Talk)**
- HEP 2020 - 38th Conference on Recent Developments in High Energy Physics and Cosmology, September 2021, Athens, Greece. **Talk**
- 16th Marcel Grossmann Meeting - MG16, July 2021, Rome, Italy, **(Talk)**
- 19th online Conference in the String Phenomenology Conference Series, June 2020, Boston, United States.
- 9th Korea Institute of Advanced Study (KIAS) Workshop on Cosmology and Structure Formation webinar, November 2020, Seoul, South Korea
- Workshop on Quantum Fields and Nonlinear Phenomena, SEENET-MTP-CERN-ICTP Joint Meeting, 2020, Romania, **(Talk)**
- SEENET-MTP Balkan School on High Energy and Particle Physics: Theory and Phenomenology, 2019, Ioannina, **(Talk)**
- HEP 2019 - Recent Developments on High Energy Physics and Cosmology, 2019, Athens **(Talk)**

## Research Interests

Theoretical Cosmology, Observational Cosmology, Machine Learning, Hubble Tension, Modified Gravity, Dark Energy, Data Analysis, Gravitational waves

## Programming Knowledge

- Python, Julia, C/C++, Fortran, HTML, Tensorflow
- COSMOMC/CAMB, MontePython/Class, Mathematica, Matlab

## Skills

Cosmology, Theoretical Astrophysics, Machine Learning Techniques, Computational Physics, General Relativity, Data Analysis

## Student Supervision

**MSc. Students** Universidad Autónoma de Madrid, UAM

1. Raul Rodriguez
2. Victor de Mora
3. Jose Javier Ortega

**BSc. Students** University of Oviedo & Ioannina

1. Alberto Nieto
2. Raul Enrique Pariente